

Summary

- Gaussian mutations are natural for continuous ESs; when turning to unbounded integer spaces, **Double-Geometric (DG) mutations obtain maximal entropy [1]**.
- Natural Evolution Strategies** provide a principled way to adapt parameters.
- The ℓ_1 -norm is the natural measure of displacement on $\mathbb{Z}^n$, and the DG distribution is its canonical mutation operator [2]. The sufficient statistic for this family gives a natural-gradient signal for **step-size adaptation**.
- Cumulating this signal through an evolution path yields the $(1, \lambda)$ -INES.
- INES learns meaningful coordinate-wise search scales and performs particularly well in **higher dimensions**, especially on Ellipse-like landscapes.

Research question. Can integer step-size adaptation be derived from first principles using a native ℓ_1 -norm mutation model and information geometry?

A natural integer step size

For coordinate-wise DG mutation, define the step size through the **expected absolute displacement**:

$$\delta_i := \mathbb{E}[|Z_i|], \quad S := \mathbb{E}[\|z\|_1] = \sum_{i=1}^n \delta_i.$$

The DG parameter q_i maps directly to this quantity:

$$\delta(q_i) = \frac{2q_i}{1 - q_i^2} \iff q_i(\delta_i) = \frac{\delta_i}{\sqrt{1 + \delta_i^2} + 1}.$$

Key point: the same quantity that measures integer step length, $|Z|$, is also the **sufficient statistic** of the search distribution.

Can we accumulate the signal?

Yes. Fading memory is well motivated for online natural-gradient learning [3]; here, it provides a momentum-like adaptation of the natural-gradient signal.

$$\phi_i \leftarrow (1 - c)\phi_i + \sqrt{c(2 - c)} g_i, \quad g_i = \frac{|z_{s,i}| - \delta_i}{\max\{\delta_i \sqrt{1 + \delta_i^2}, 1\}}.$$

We **prove** that this update is a fading-memory online natural gradient estimator.

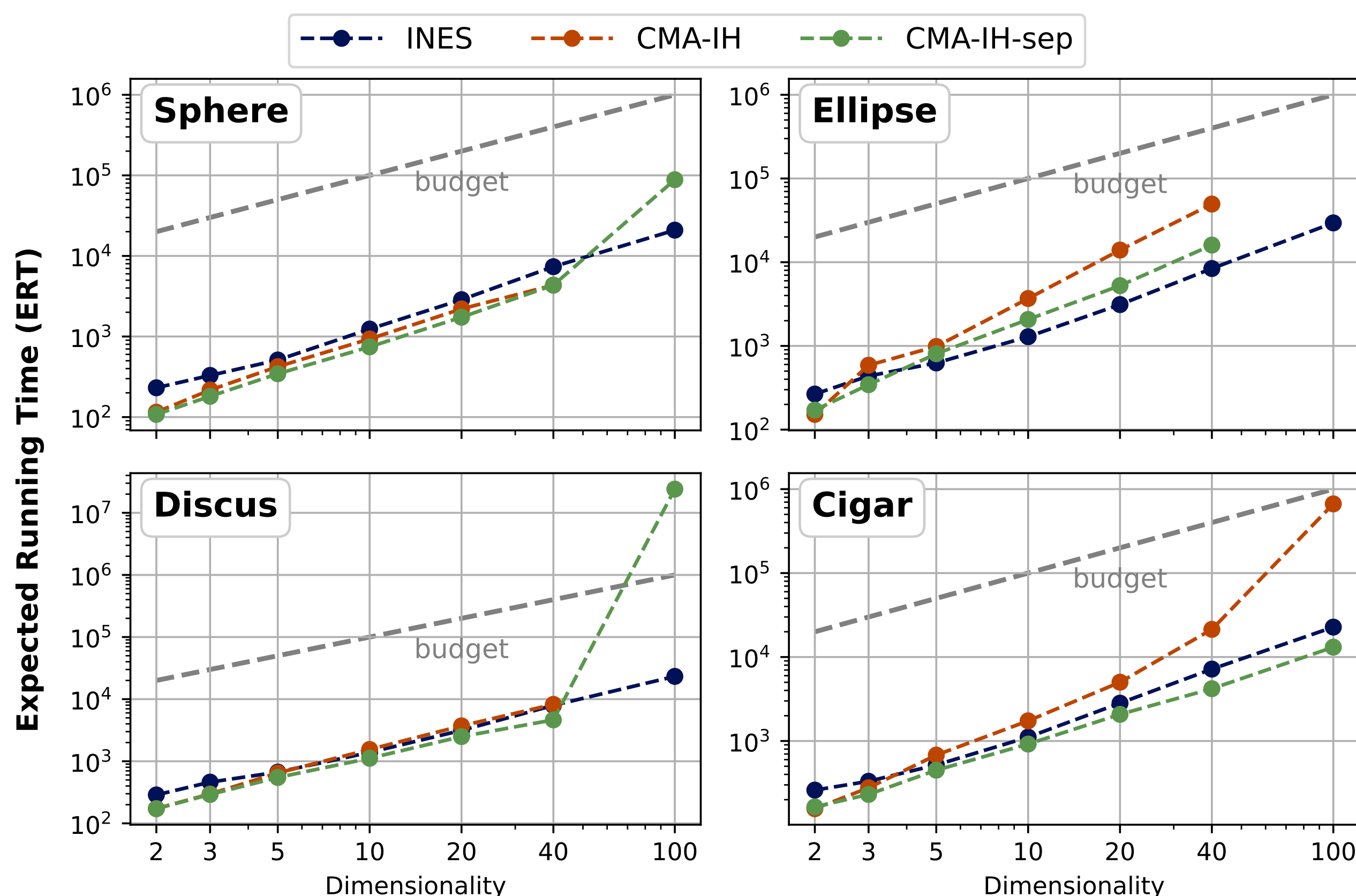
A concrete $(1, \lambda)$ -INES

- For $k = 1, \dots, \lambda$: sample $z_k \sim \bigotimes_i \text{DG}(p_i(\delta_i))$ and evaluate $f(\mathbf{m} + z_k)$.
- Select $s = \arg \min_k f(\mathbf{m} + z_k)$ and set $\mathbf{m} \leftarrow \mathbf{m} + z_s$.
- Compute the Fisher-normalized signal g_i and update the cumulative path ϕ_i .
- Adapt the expected absolute step size: $\delta_i \leftarrow \delta_i \exp(\eta \phi_i)$.

Calibration in the paper: $\lambda = 10$, $c(n) = 1 - 1.5/n$, and $\eta(n) = (2/n)^{1/3}$.

Benchmarking on Unimodal Quadratic functions

25 runs per function/dimension, budget $10^4 \cdot n$; lower ERT is better. Baselines are CMA-IH and CMA-IH-SEP.



Continuous vs. Integer Mutation

Continuous search

$Z \in \mathbb{R}$, $Z \sim \mathcal{N}(0, 1)$
 ℓ_2 geometry uses squared displacement:
 $\|z\|_2^2 = z^2$.
 The maximum-entropy density is
 $p(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right)$.

Integer search

$Z \in \mathbb{Z}$, $Z \sim \text{DG}(q)$
 ℓ_1 geometry uses absolute displacement:
 $\|z\|_1 = |z|$.
 The maximum-entropy mass function is [1]:
 $p(z) = \frac{1 - q}{1 + q} \exp(\log(q) |z|)$.

Same principle, different geometry: $z^2 \longrightarrow |z| \Rightarrow \mathcal{N}(0, 1) \longrightarrow \text{DG}$

From DG to the natural gradient

In exponential-family form,

$$\theta_i = \ln q_i, \quad \phi(z_i) = |z_i|,$$

with

$$A'(\theta_i) = \delta_i, \quad A''(\theta_i) = \text{Var}[|Z_i|].$$

For a selected mutation $z_{s,i}$, the score is therefore

$$\nabla_{\theta} \mathcal{L} = |z_{s,i}| - \delta_i.$$

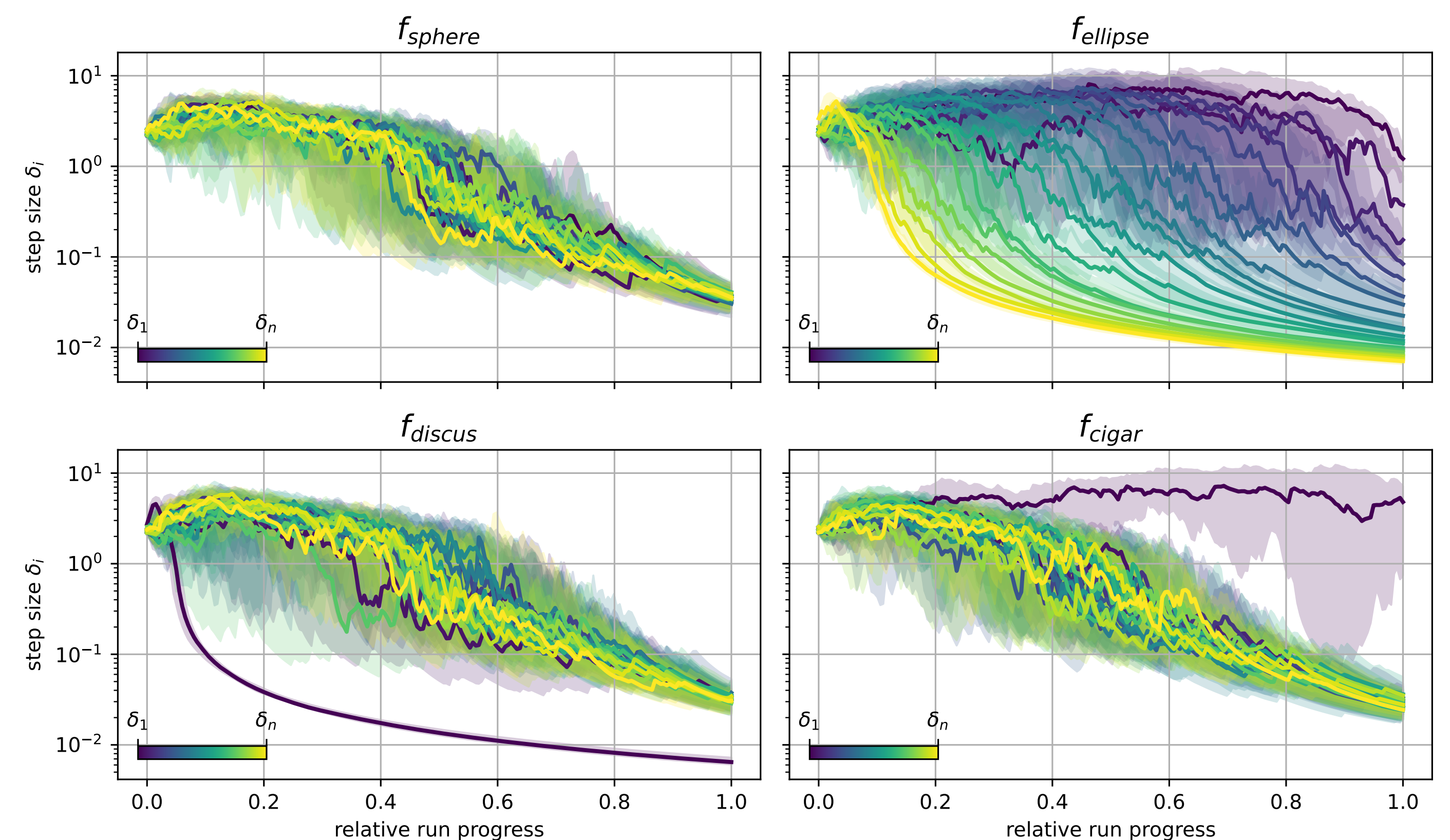
A step larger than expected gives a positive signal; a smaller step gives a negative signal. Since the Fisher information is

$$\mathcal{F}(\theta_i) = \text{Var}[|Z_i|],$$

the natural gradient becomes

$$\widetilde{\nabla}_{\theta_i} = \frac{|z_{s,i}| - \delta_i}{\text{Var}[|Z_i|]}.$$

What does INES learn?



Sphere: coordinates evolve similarly.

Discus: the distinguished coordinate is suppressed.

Ellipse: step sizes separate according to conditioning.

Cigar: the distinguished coordinate is amplified.

The learned step sizes behave as expected from the benchmark geometry, supporting the interpretation of δ_i as meaningful coordinate-wise scales.

Conclusions & future work

- The DG distribution gives a **lattice-native** mutation model for integer ES.
- A natural-gradient signal for step-size adaptation is provided by the sufficient statistic.
- Cumulation turns this signal into a stable $(1, \lambda)$ -INES update.
- INES is most compelling on **Ellipse** and remains robust in high-dimensional settings.

Current limitations / next steps

No explicit global step-size component; coordinate-wise adaptation only. Natural extensions are multi-parent recombination, a global scale, and modeling dependencies between integer variables.

References

- [1] Günter Rudolph. An evolutionary algorithm for integer programming. In Yuval Davidor, Hans-Paul Schwefel, and Reinhard Männer, editors, *PPSN III*, pages 139–148. Springer Berlin Heidelberg, 1994.
- [2] Ofer M. Shir and Michael Emmerich. Foundations of correlated mutations for integer programming. In *FOGA 18, FOGA'25*, pages 285–296. Association for Computing Machinery, 2025.
- [3] Yann Ollivier. Online natural gradient as a kalman filter. *Electronic Journal of Statistics*, 12(2):2930–2961, 2018.